Chapter 10

Powders With Quasicrystalline Structure

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Chapter Outline

DISCOVERY AND STRUCTURE OF QUASICRYSTALLINE MATERIALS

Quasicrystals (QC) are solids (generally, there are intermetallic phases) that possess a unique crystallographic structure characterized by long-range aperiodic order and crystallographically forbidden rotational symmetries. Accordingly, they exhibit five-fold, seven-fold, eight-fold, 10-fold, 12-fold, 15-fold, etc., rotational symmetries and a quasiperiodic translational order. For the discovery of quasicrystals, D. Shahtman was awarded the Nobel Prize in Chemistry in 2011. QCs were discovered in a rapidly solidified $Al_{0.86}Mn_{0.14}$ alloy in 1984 [1]. The material was obtained after rapid solidification from the liquid state at a cooling rate up to 1 million degrees per second. The samples of such alloys dissipated the electron beam and formed a clear diffraction pattern with a five-fold symmetry (Fig. 10.1). Many scientists consider the discovery of QC (along with metallic glasses, high-temperature superconductors, and nanocrystalline materials) as the most important achievements of material science in the 20th century.

Despite the relatively recent discovery, QCs gradually keep their place among the materials with a wide range of unique properties that are used in industry.

The structure of the QC is based on the icosahedron geometric figure, unlike most crystals, which have structures based on simple geometric figures such as a cube, tetrahedron, and octahedron. However, the icosahedron has the axis of the fifth order and the complete filling of space by these figures is impossible. In addition, the icosahedron that consists of 13 atoms, one of which is situated in the center and 12 of which are located above the sphere surface, is imperfect because there are small clearances between the 12 atoms on the surface. The addition of new atoms increases the imperfection. Therefore, the QCs usually are formed by icosahedral clusters with agglutinant atoms between them. It is shown that the structure of the icosahedral QC is formed by clusters of about 1.0 nm in dimension (0.9 nm in the case of Al-Pd-Mn [2]). The models of such clusters consist of 51 atoms for Al- Pd—Mn [3] and 33 atoms for Al-Cu-Fe [4].

Soon after the discovery of the icosahedral QC structure, the decagonal QC structure with a 10th-order symmetry axis was found in 1985 [5]. The octagonal QC with eighth-order symmetry axes and the dodecagonal QC with 12th-order symmetry axes also exist. The Al-based QCs are the most known and studied; however, the QCs also exist on V, Nb, Mn, Zn, Ti, Zr, Mg and other metal base. The majority of the discovered QCs are manufactured by rapid crystallization and are metastable. Currently more than 200 metastable QCs are known. A number of stable QC phases (in systems of Al-Cu-Fe, Al-Pd-Mn, Al-Cu-Co, Al-Co-Ni, and other alloys) also exist and are presented in equilibrium diagrams; they can be manufactured by the slow cooling rate of the melt [6,7].

Properties and Application of QC Materials

The electronic structure of a QC is usually characterized by the neighborhood of the Fermi surface to a minimum on the curve of the density states. In connection with this, the QC interatomic bonds have a mainly covalent character that accounts

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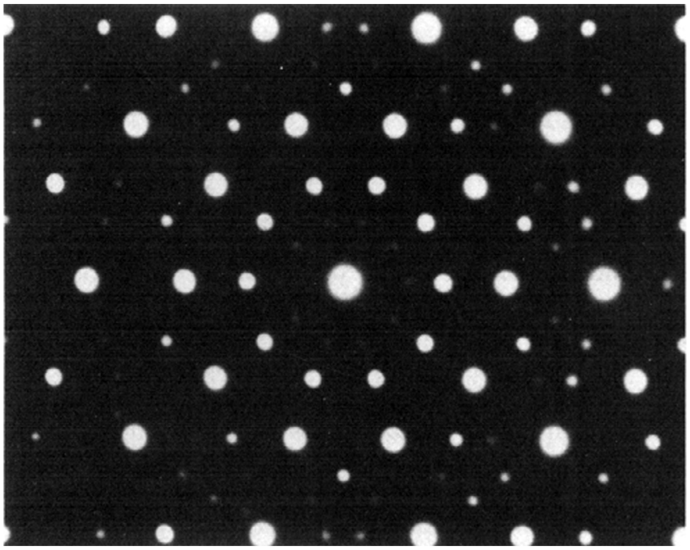

FIG. 10.1 Electron diffraction pattern of five-fold symmetry from the rapidly solidified alloy Al_6Mn.

for the existence in QCs of a wide energetic pseudogap. Accordingly, a small number of free electrons causes their low electric and thermal conductivity. This sets conditions for the specific and unique nature of the QC's physical and chemical properties. Thus, in annealed Al-Cu-Fe quasicrystals, the specific electric resistance achieves by the temperature 0.4 K the value of $\rho_{0.4K} = 0.01\ \Omega$ cm that is the level of the electric resistivity of semiconductors. In contrast to the usual conditions of crystals, the electric resistivity of OC by the increase of temperature is decreased; for example, for Al-Cu-Fe and Al-Pd-Mn quasicrystals, the ratio values ρ_{4K}/ρ_{100K} in the range of ~2–3 (where ρ_{4K} and ρ_{300K} are the electric resistivity at temperature 4 K and 300 K, respectively) are typical [8]. The thermal conductivity of the Al-Cu-Fe quasicrystals is of 1–3 $W\,m^{-1}\,K^{-1}$ that is on the level of the thermal conductivity of oxide ceramics ZrO_2 [9].

The high level of mechanical properties and deformation characteristics of QCs single them out into a special class of materials. Direct study of the QC's deformed structure by transmission electron microscopy (TEM) shows that the deformation of QCs has a dislocation character [10,11]. Dislocations in the QCs have the phonon and phason components of distortions. Phason defects can move only by diffusion. That, along with a high value of Peierls stress (that is inherent in covalent crystals), results in a high value of yield strength and hardness. The hardness of different QCs is in the range of 4–11 GPa, and Young's modulus is within 80–200 GPa due to the high level of binding forces in the QC. However, the fracture toughness of QC is $K_{1c} = 0.5\text{–}1.5\ MPa\,m^{1/2}$, which is close to the fracture toughness of silicon and a number of ceramic materials.

The unique properties of QC materials make possible a wide application of QCs in different industries. The main directions of application of QC materials are shown in Table 10.1.

MANUFACTURING TECHNIQUE OF QC POWDERS

Mechanical Alloying

Mechanical alloying is a technique to synthesize both equilibrium and nonequilibrium phases of materials starting from elemental powders. Mechanical alloying is a completely solid-state processing technique and therefore limitations imposed by the phase diagrams do not function here. More detailed information of mechanical alloying is contained in Chapter 3.

QC powders have been synthesized by mechanical alloying since 1989 [12,13]. The value of the interdiffusion coefficient determines the time formation of the QC phase. For example, the formation of the icosahedral phase in the Mg-Al-Zn system requires only 3–10 min [14], whereas in the Ti-Zr-Ni system it requires approximately 2 h [15], which is associated with a lower diffusion rates of elements in the system Ti-Zr-Ni. The types of powders with icosahedral QC structure and the conditions for their making are summarized in Table 10.2. In some cases, the subsequent annealing is required to form the QC structure in powders with an amorphous or nanocrystalline structure.

Recently, the synthesis of the decagonal QC phase in the Al–Cu–Cr system by the mechanical alloying technique has been reported [16]. This phase has grains with a faceted shape. The above composition is in the range of 71–73 at.% Al, 11–12 at.% Cu, and 15–18 at.% Cr.

TABLE 10.1 Application Area of QC Materials

Areas of Application	Properties	QC Composition
Thermal barriers (aviation turbine and turbine generators, diesel engines, automotive industry)	Low thermal conductivity	Al-Cu-Fe, Al-Co-Cr-Fe
Disperse-reinforcing particles in aluminum alloys, steels, and composites	Hardness and high mechanical properties at elevated temperatures	Al-Fe-Cr, Al-Cu-Fe, martensite aging steels
Accumulation of hydrogen	The high adsorption capacity for hydrogen	QC on Ti base
Selective light absorbers	The absorption of light radiation of certain wavelengths	Al-Cu-Fe, Al-Cu-Fe-Cr;
Nonburn coating for chemical reactors, kitchen utensils, and others.	Corrosion resistance, high hardness, low surface energy (no sticking)	Al-Cu-Fe-Cr
Coatings for tools	Wear resistance, low coefficient of friction	Al-B-Cu-Fe, Al-Pd-Mn
Catalysts in chemical reactions	The catalytic activity	Al-Pd, Al-Pd-B Al-Pd-Fe, Al-Pd-Mn, Al-Pd-Cr, Al-Cu-Fe, Al-Cu-Co, Al-Pd-Co

Vaporization by Laser Beam

A technique for manufacturing powders containing an icosahedral phase by evaporating by laser beam of ribbons from Al-Cu-Fe and Al-Pd-Mn system alloys of 50 μm in thickness is described in the work [17]. The laser with the $\lambda = 248$ nm wavelength was used. Targets of 2.5 cm in diameter were placed in a vacuum chamber filled by high purity argon at a pressure of 10 mmHg. The laser beam was focused on a spot of 0.5–1 mm in diameter on the surface of a rotating target. Pulses of 20 ns duration were emitted with a frequency of 10 Hz. The pulse energy (about 50 mJ/pulse) remained constant on this level during all experiments. Evaporated powders were collected in the nylon filter of the vacuum system. Powder produced by laser evaporation had a flake shape, a 10–20 μm size range, and a superfine grain structure.

The initial alloy was made from a mixture of high-purity Al, Cu, and Fe in accordance with the composition $Al_{62}Cu_{25.5}Fe_{12.5}$ and following evaporation-condensation in the argon atmosphere. Studying the manufactured powder had shown the presence of the icosahedral QC phase with an admixture of the crystalline β phase. The powders of the $Al_{70}Pd_{21}Mn_9$ alloy containing the QC phase were also produced by this technique [18].

Crushing of Ingots

For the laboratory investigation of the QC alloys' properties, the small ingots (50–100 g) were manufactured by melting in a high-frequency furnace in an argon environment followed by the melt pouring into the cooled copper crucible. The samples of several mm in diameter and of several cm in length were formed by drawing the melt. The crystallization rate of 10 mm in diameter samples was about 250 K/s. Thus, the ingots of 47 alloys based on Al-Cu-Fe, Al-Co-Fe, Al-Co-Ni, and other systems were produced. Such ingots contain, as a rule, several phases, and the QC phase in them was elevated by 24 h annealing at 1085 K that allowed the increase of the QC phase content in the ingot up to 95% [19]. The ingot obtained in an induction furnace under an argon protective atmosphere was crushed to produce a powder with a size in the range 20–50 μm, which was used later for sintering under uniaxial pressure [20].

Evidently, producing powders by crushing such ingots on an industrial scale is not expedient. In work [20], the QC powders were produced by comminution of the 5 kg ingots (50 mm diameter) and the coatings from these powders were investigated. Crushing was carried out in a mechanical agate mortar. Shevchukov et al. [16] reported the technology for production from 100 kg melt the 2 kg Al-Cu-Fe ingots contain 95 wt% of QC phase. The ingots are crushed in a tumbling ball mill with steel balls. This technology can be a base for commercial production of QC alloys by crushing ingots.

A typical powder particle image of a crushed QC ingot is shown in Fig. 10.2. Along with elongated particles and particles with sharp edges, the powder contains many particles of a polyhedral shape. Evidently, it is due to a high degree of symmetry in the icosahedral phase.

TABLE 10.2 Icosahedral QC Phases Formed after Mechanical Alloying and Rapid Solidification

Chemical Composition	Type of Mill	Ball-to-Powder Ratio	Milling Intensity	Milling Time (h)	Subsequent Annealing	Ref.
$Al_{65}Cu_{20}Co_{15}$	Fritch P5	15:1	–	55	873 K, 1 h	17
Al-Cu-Cr	Fritch P5	15:1	–	–	–	17
$Al_{70}Cu_{20}Fe_{10}$	Fritch P5	10:1	–	40	–	18
$Al_{65}Cu_{20}Fe_{15}$	Fritch P7	15:1	–	15	–	19
$Al_{65}Cu_{20}Fe_{15}$	AGO-2 U	15:1	–	15	823 K	20
Al_{70-x} (Cu, Fe)$_{30+x}$	Fritch P5	10:1	–	20–30	–	21
$Al_{65}Cu_{20}Mn_{15}$	Fritch P5	15:1	–	90	–	19
$Al_{65}Cu_{20}Mn_{15}Ge_{25}$	Fritch	20:1	450–650 rpm	66	–	22
$Al_{75}Cu_{15}V_{10}$	Fritch P5/P7	15:1	–	–	723 K, 1 h	23
$Al_{92}Mn_6Ce_2$	Fritch P5	13:1	300 rpm	700	–	23
$Al_{91}Mn_7Fe_2$	Fritch P5	13:1	300 rpm	700	–	23
$Al_{50}Mn_{20}Ge_{30}$	Fritch	20:1	45–650 rpm	47	–	24
$Al_{50}Mn_{20}Si_{20}Ge_{10}$	Fritch	20:1	45–650 rpm	80	–	24
$Al_{70}Pd_{20}Mn_{10}$	Fritch P7	15:1	–	30	–	25
$Mg_{32}Cu_8Al_4$	Planetary	–	900 rpm	–	–	13
Mg-Al-Pd	Planetary mill	40:1	150 G	4	–	26
Mg_{32} (Al, Zn)$_{49}$	Fritch	–	600 m/s^2	10 min	–	15
$Mg_3Zn_{5-x}Al_x$ (x = 2–4)	Planetary mill	40:1	900 rpm	–	–	13
$Mg_{44}Al_{15}Zn_{41}$	Planetary ball mill	–	600 m/s^2	30 min	–	27
$Ti_{56}Ni_{18}Fe_{10}Si_{16}$	SPEX 8000	6:1	–		1023 K, 30 min	28
$Ti_{45}Zr_{38}Ni_{17}$	Fritch P7	8:1	9.5	–	828 K	29
$Ti_{45}Zr_{38}Ni_{17}$	AGO-2	20:1	400 m/s^2	2	–	16

Centrifugal Atomization

This technique is an intermediate between laboratory and commercial production ones. The centrifugal atomization was used [21] for manufacturing Al-Cu-Fe powder with a QC component. The several hundreds grams ingot previously melted in an induction furnace in argon environment was remelted in a crucible placed in the center of the atomization chamber in a helium atmosphere. After achieving the pouring temperature (exceeding the melting point up to 200 K), the crucible's bottom drain aperture was opened, and the jet melt fell on the disk rotating with a speed of 30,000 rpm. The melt spread over the disk under centrifugal force is atomized with a generation of fine melt droplets, which are solidified during their flight to the chamber walls. However, the melt cooling rate by centrifugal atomization does not exceed 10^3 K/s. Therefore, the QC phase content in powder was comparatively small (about 30%). Further powder annealing did not bring it to the one-phase condition.

The powders of AI-Mn, Al-Cr, and AI-Mn-Cr system alloys with the QC phase were made using an apparatus in which the molten alloy was first atomized into small droplets by a nozzle atomizer, then the droplets collided with a high-speed rotating disc, and finally were atomized. Therefore, the droplets were rapidly solidified. The calculated cooling rate was in the 10^5–10^7 K/s range [22].

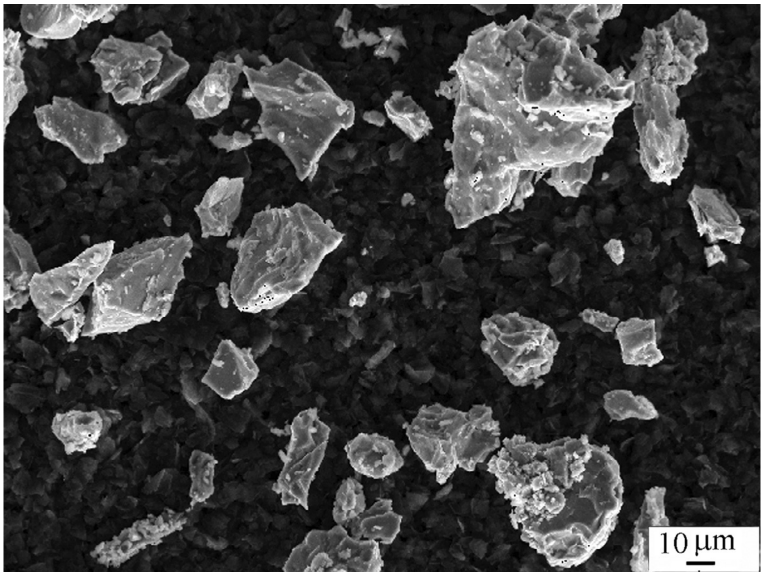

FIG. 10.2 SEM micrograph of $Al_{63}Cu_{25}Fe_{12}$ alloy powder produced by crushing ingots.

Gas-Atomized Powders

At present, gas atomization (GA) is the main commercial production method of aluminum and aluminum alloys and is described in the Ref. [23]. The cooling rate by GA achieves 10^5 K/s, which is about three orders higher than the cooling rate of ingot in a water-cooled copper mold. A high solidification rate provides fineness structure and a more homogeneous composition in powders.

The technique of manufacturing QC Al-Cu-Fe alloys by inert GA is presented in the patent [24]. As reported [25], in Al-Cu-Fe powder produced by GA, an x-ray investigation reveals two phases: the icosahedral QC phase and the cubic crystalline (β). Very small crystals of the crystalline λ-phase are revealed only by TEM investigation. Inert GA protects the surface of powder particles from strong oxidation. Thus, in the $Al_{63}Cu_{25}Fe_{12}$ powder produced by argon atomization, the content of oxygen varied from 0.205 to 0.065 wt% for powder size fractions of 0–25 μm and 75–106 μm, respectively [24]. The oxygen content determined by the Leco 12 technique of melting in an inert gas showed it lowering with the increase of particle size due to reducing the specific surface of particles. The part of the QC phase in these fractions varied from 69% to 55% because the solidification rate in fine particles is higher.

In comparison with ingot-crushed powders, which have an irregular splinter particle shape, GA powders have a spherical shape. The spherical QC powders, produced by GA of the $Al_{65}Cu_{20}Cr_{15}$ [25] alloy, are shown in Fig. 10.3.

For coatings, subject to size fractions of QC GA powders, different types of plasma torches are used. For the coating by the 10–25 μm size fraction, minitorches are used. For the size fraction 25–45 μm or even 45–75 μm, conventional torches are used. Small particles below 10 μm in diameter are not used because they evaporate in the high temperature plasma while the particles with size above 75 μm do not stay long enough within the flame to reach full melting [26].

The $Al_{65}Cu_{20}Fe_{15}$ QC spherical shape powder with a median diameter of 17.4 μm was made by argon atomization using Rayleigh-plateau forces in primary mode [26]. The Nanoval system, which used an atomizer with a Laval nozzle, was applied. Powder consisted of the icosahedral QC phase dendrites and a small amount of Al(Cu, Fe) cubic phase in the interdendritic areas.

The French firm CRISTOME (Coating solutions Saint-Gobain) manufactures gas-atomized QC powders. Commercially available powders include CRISTOME A_1 ($Al_{71}Fe_{8.7}Cr_{10.6}Cu_{9.7}$), CRISTOME F_1 (Al_{59} $Fe_{12.4}$ $Cu_{25.5}$ B_3), and CRISTOME BT_1 ($Al_{71.3}Fe_{8.1}Co_{12.8}Cr_{7.8}$) [27]. The main properties of CRISTOME QC powders are summarized in Table 10.3.

The new application of gas-atomized QC Al-Cu-Fe-B alloy powders was conductive to a commercial development of polymer-matrix composites reinforced with QC particles [28]. The process is based on selective laser sintering (SLS), which is one of the three-dimensional additive manufacturing technologies used in the mechanical industries. For example, the intake manifold from the polyamide composite reinforced by QC particles was made by SLS [29].

As reported [30–33], the gas atomized Al-Fe-Cr alloy powders strengthened by nanoquasicrystalline particles are produced. Various gases are used for atomization: helium [31]; nitrogen [32] under 0.45 and 0.5 MPa pressure, respectively; and argon [30] under 9.8 MPa pressure. Argon was used for the atomization of a number of Al-Fe-Cr-Nb system alloys [33].

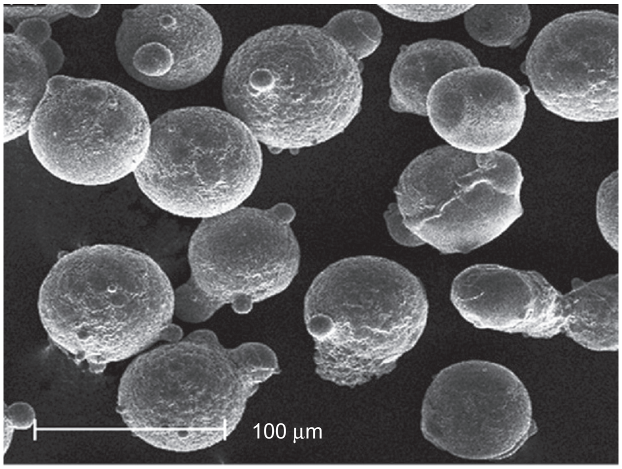

FIG. 10.3 SEM micrograph of 61–74 μm fraction of QC $Al_{65}Cu_{20}Cr_{15}$ powder, produced by argon atomization.

TABLE 10.3 Main Properties of CRISTOME Quasicrystalline Powders

Compositions (at.%)	Al_{71} $Fe_{8.7}$ $Cr_{10.6}$ $Cu_{9.7}$	Al_{59} $Fe_{12.4}$ $Cu_{25.5}$ B_3	$Al_{71.3}$ $Fe_{8.1}$ $Co_{12.8}$ $Cr_{7.8}$
Solidus, K	1163	1069	1364
Liquidus, K	1309	1343	1393
Thermal conductivity (W/mK)			
20 K	2	3	2
300 K	3.2		
800 K	6	7.5	5
Thermal diffusivity (mm^2/s)			
20 K	0.80	1	1.45
300 K	1.10		
800 K	2.39		
Electrical resistivity (μΩ cm)	600		400
Thermal dilatation coefficient ($\times 10^{-6}$ K^{-1})	15.5–18.6	15.4–23.5	13.7–15
Density (g/cm^3)	3.99	4.72	4.09

Water-Atomized Powders

A well-known technique of melt atomization by jets of high-pressure water (WA) enables increasing the melt cooling on a single order in comparison with GA. This process is widely used for producing powders of ferrous metals and some non-ferrous metals (e.g., copper and alloys on its base, nickel, nickel alloys). However, this process was not applied to manufacturing aluminum and aluminum alloy powders for two main reasons: First, due to the explosion hazard that arises from the emanation of hydrogen as a result of powder interaction with water, and, second, for fear of powder oxidation and the deterioration of powder properties. The problem of powder quality and safety of the WA atomization process to produce Al alloy powders has been recently solved at the Institute for Problems of Materials Science (Kiev, Ukraine) [34]. The developed solution has been realized in the form of a pilot plant and is based on the following technical innovations: using for atomization water with the properties of a weak electrolyte, monitoring its temperature and hydrogen index, the hydraulic classification of powders in particle size, a contact-convection drying process with a high drying efficiency for drying chemically active substances, and preventing powder oxidation and thermal self-ignitions.

WA-atomized powders have an irregular shape and uneven surfaces (Fig. 10.4). As follows from direct TEM observations, the surface oxide of water-atomized powders is nonuniform in its thickness. Coarse surface oxide islands, with a typical thickness of 30–40 monolayers, cover 30%–70% of the particle's surface. The rest is covered with a thin oxide film of about 2 to a maximum of 5–6 monolayers [35]. The cooling rate of melt is 10^6 K/s and higher, which is necessary for obtaining QC phases.

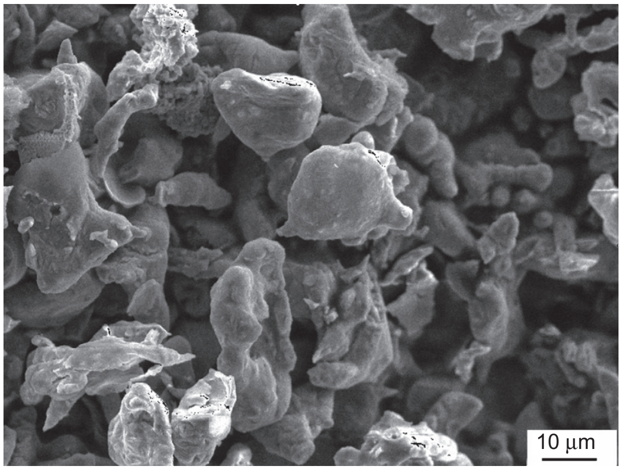

FIG. 10.4 SEM micrograph of 40–63 µm fraction of QC $Al_{63}Cu_{25}Fe_{12}$ powder, produced by water atomization.

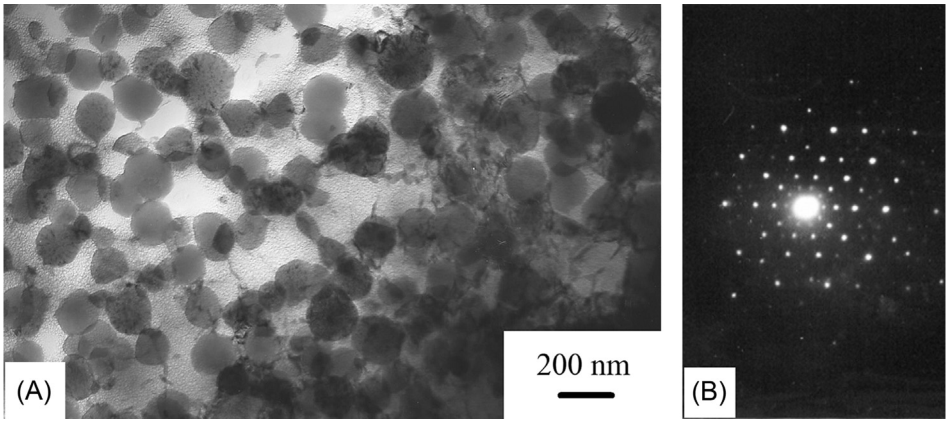

FIG. 10.5 TEM image of water-atomized $Al_{96}Fe_{2.5}Cr_{2.5}Ti_1$ alloy powder (A) and electron diffraction pattern of five-fold symmetry from a quasicrystalline particle (B).

A polygonal cell structure of the Al matrix with a cell size of 500 nm and nanoquasicrystalline strengthening particles with a size of 200–300 nm is observed in the $Al_{96}Fe_{2.5}Cr_{2.5}Ti_1$ alloy, obtained by extrusion of water-atomized powder (Fig. 10.5A) [36]. The TEM image illustrates an electron diffraction pattern of five-fold symmetry from the QC particle (Fig. 10.5B).

Among wrought aluminum alloys, the Al-Fe-Cr alloys based on the WA powders possess uniquely high mechanical properties both at room and elevated temperatures (Table 10.4). Such high strength ensues due to a high hardness and high Young modulus of the QC while plasticity is provided by the special deformation treatment of nanoquasicrystals.

The QC based on WA powders technique used for manufacturing $Al_{63}Cu_{25}Fe_{12}$ powder [37] results in obtaining two-phase powder (QC + crystalline β) (Fig. 10.6A), following conversion to the QC phase by vacuum annealing at 973 K over 2 h (Fig. 10.6B). For producing $Al_{64}Cu_{25}Fe_{12}$ powder aluminum A5 (99.95%), cathodic copper (99.99%) and iron "Armco" (99.92%) were used. The charge composition consists of (43Al-40.1Cu-16.9Fe) wt%. The overheating of pouring temperature was 200–250 K. The 6 mm melt stream flowing from the crucible was atomized by water jets under 100 MPa pressure. Dried powders were separated by sieving into the following size fractions: 0–63, 63–100, 100–160, 160–200, and >200 µm.

QC Al-Cu-Fe coatings, especially with additional alloying, have low thermal conductivity (1.65–2.42 $W\,m^{-1}\,K^{-1}$) combined with the thermal expansion coefficient (14.5×10^{-6}–$18.5\times10^{-6}\,K^{-1}$ at a temperature of 373–973 K) approximate to the thermal expansion coefficient for structural materials of nickel, aluminum, and iron base. Therefore, one can recommend them for use as thermal barriers for gas turbines, internal combustion engines, etc.

Water-atomized QC Al-Cu-Fe system powders were also used to produce the surface-hardened composite layers on an aluminum alloy by ultrasonic impact processing [38].

TABLE 10.4 Mechanical Properties of Extruded Rods from Al-Fe-Cr WA Powders With Nanoquasicrystalline Strengthening

Chemical Compositions (at. %)	Test Temperature					
	293 K			573 K		
	[a]YS (MPa)	[b]UTS (MPa)	[c]El (%)	YS (MPa)	UTS (MPa)	El (%)
$Al_{96}Fe_3Cr_3$	485	542	7.0	283	297	3.5
$Al_{96}Fe_{2.5}Cr_{2.5}Ti_1$	546	585	8.4	328	345	3.9
$Al_{96}Fe_{2.5}Cr_{2.5}Ti_{0.5}Zr_{0.5}$	648	677	7.0	331	351	1.8

[a]*Yield strength.*
[b]*Ultimate tensile strength.*
[c]*Elongation.*

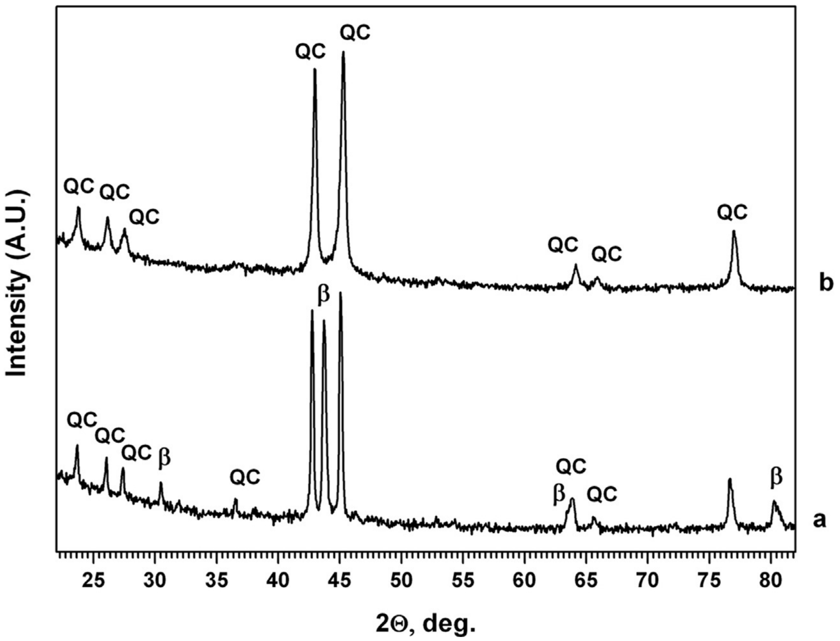

FIG. 10.6 X-ray patterns of water-atomized $Al_{63}Cu_{25}Fe_{12}$ powder of 40–63 μm fraction before (a) and after (b) vacuum annealing at 973 K over 2 h.

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